

## POPE AFB, NC, USA

WMO: 723030

Lat: 35.174N

Lon: 79.009W

Elev: 218

StdP: 14.58

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13714

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 20.5       | 24.5 | 3.3                           | 6.6 | 29.1 | 8.9 | 8.7 | 34.2 | 23.2                  | 55.3 | 19.7 | 54.3 | 3.7                   | 270  | 0.415 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 18.6                   | 96.8            | 76.1 | 94.3 | 75.6 | 91.7 | 75.0 | 79.8                | 88.9 | 78.5 | 87.8 | 77.6 | 87.0 | 8.3                  | 240  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 77.5                            | 144.1 | 82.8 | 76.4 | 138.7 | 82.1 | 75.1 | 132.6 | 81.3 | 43.4          | 89.3 | 42.1 | 88.0 | 41.0 | 87.3 |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |      |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |      |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max  |       |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p)  |       |
| 19.2              | 16.9 | 14.8 | DB                         | 12.8 | 99.9               | 4.9 | 2.6                                                | 9.3 | 101.8      | 6.4 | 103.4      | 3.7 | 104.8      | 0.1  | 106.7 |
|                   |      |      | WB                         | 11.7 | 81.8               | 4.8 | 2.3                                                | 8.2 | 83.4       | 5.4 | 84.8       | 2.7 | 86.0       | -0.8 | 87.7  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 62.9   | 42.8  | 46.6  | 53.4  | 63.4  | 71.5  | 78.7  | 81.2  | 79.8  | 73.9  | 63.5  | 52.7  | 46.6 |
|                                                                               | DBStd       | 15.62  | 10.58 | 9.70  | 9.58  | 8.43  | 6.92  | 5.28  | 4.07  | 4.54  | 5.59  | 7.79  | 8.59  | 9.37 |
|                                                                               | HDD50       | 773    | 273   | 165   | 76    | 6     | 0     | 0     | 0     | 0     | 0     | 3     | 68    | 182  |
|                                                                               | HDD65       | 2823   | 689   | 519   | 376   | 130   | 24    | 1     | 0     | 0     | 5     | 124   | 381   | 574  |
|                                                                               | CDD50       | 5488   | 51    | 70    | 182   | 407   | 666   | 860   | 967   | 924   | 718   | 420   | 147   | 76   |
|                                                                               | CDD65       | 2064   | 2     | 4     | 18    | 81    | 225   | 410   | 502   | 459   | 273   | 76    | 10    | 4    |
|                                                                               | CDH74       | 20600  | 11    | 30    | 178   | 835   | 2207  | 4302  | 5358  | 4693  | 2244  | 657   | 67    | 18   |
|                                                                               | CDH80       | 8771   | 1     | 2     | 28    | 237   | 849   | 2007  | 2519  | 2141  | 819   | 163   | 4     | 1    |
| Wind                                                                          | WSAvg       | 6.1    | 6.1   | 6.6   | 7.2   | 7.7   | 6.8   | 6.2   | 5.9   | 5.3   | 5.7   | 5.3   | 5.2   | 5.5  |
| Precipitation                                                                 | PrecAvg     | 45.4   | 3.3   | 2.9   | 3.8   | 3.1   | 3.1   | 4.7   | 5.1   | 5.0   | 5.0   | 3.0   | 3.2   | 3.0  |
|                                                                               | PrecMax     | 61.7   | 6.9   | 6.1   | 7.0   | 6.4   | 7.2   | 12.3  | 10.0  | 10.0  | 15.2  | 7.7   | 9.0   | 5.7  |
|                                                                               | PrecMin     | 33.3   | 0.9   | 1.1   | 1.1   | 0.9   | 1.0   | 0.4   | 1.5   | 1.3   | 1.4   | 0.0   | 0.9   | 1.1  |
|                                                                               | PrecStd     | 6.2    | 1.7   | 1.3   | 1.4   | 1.5   | 1.6   | 2.9   | 2.2   | 1.8   | 3.1   | 2.0   | 1.9   | 1.2  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 74.9  | 78.4  | 83.8  | 90.4  | 95.0  | 99.8  | 99.5  | 99.6  | 93.6  | 89.7  | 80.0  | 76.8 |
|                                                                               |             | MCWB   | 65.1  | 64.3  | 63.5  | 68.9  | 72.0  | 75.4  | 77.5  | 76.8  | 73.7  | 72.0  | 66.6  | 65.1 |
|                                                                               | 2%          | DB     | 69.9  | 73.0  | 79.0  | 85.8  | 91.1  | 95.6  | 96.4  | 95.5  | 90.5  | 84.2  | 75.3  | 71.7 |
|                                                                               |             | MCWB   | 61.4  | 60.9  | 63.1  | 66.5  | 71.4  | 75.3  | 77.5  | 76.2  | 73.7  | 70.2  | 64.0  | 64.2 |
|                                                                               | 5%          | DB     | 65.2  | 68.3  | 74.9  | 82.2  | 88.2  | 92.9  | 93.4  | 92.7  | 88.0  | 81.0  | 72.2  | 67.6 |
|                                                                               |             | MCWB   | 58.2  | 58.3  | 60.5  | 65.1  | 70.5  | 74.6  | 76.8  | 76.2  | 73.3  | 68.2  | 62.7  | 62.3 |
|                                                                               | 10%         | DB     | 61.0  | 64.1  | 70.6  | 79.1  | 84.7  | 90.2  | 90.9  | 90.0  | 85.2  | 77.4  | 68.2  | 63.4 |
|                                                                               |             | MCWB   | 54.5  | 55.8  | 58.5  | 63.7  | 69.3  | 73.8  | 76.0  | 75.3  | 72.2  | 67.1  | 61.2  | 58.0 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 67.8  | 67.1  | 69.1  | 73.3  | 76.9  | 79.4  | 82.4  | 81.4  | 78.1  | 76.1  | 71.1  | 69.0 |
|                                                                               |             | MCDB   | 72.0  | 74.0  | 75.9  | 80.2  | 86.7  | 89.9  | 91.6  | 89.0  | 86.4  | 81.7  | 75.2  | 72.2 |
|                                                                               | 2%          | WB     | 63.8  | 64.2  | 66.1  | 70.9  | 74.5  | 78.0  | 80.3  | 79.5  | 76.8  | 74.0  | 68.0  | 66.0 |
|                                                                               |             | MCDB   | 67.7  | 69.9  | 73.9  | 80.0  | 84.2  | 88.9  | 89.4  | 88.2  | 84.8  | 80.1  | 72.1  | 70.1 |
|                                                                               | 5%          | WB     | 60.2  | 61.2  | 63.9  | 68.8  | 73.0  | 76.9  | 78.9  | 78.3  | 75.6  | 71.9  | 65.3  | 62.6 |
|                                                                               |             | MCDB   | 64.3  | 67.2  | 70.4  | 77.3  | 82.6  | 87.7  | 88.3  | 87.1  | 82.8  | 77.8  | 70.0  | 66.2 |
|                                                                               | 10%         | WB     | 55.0  | 57.0  | 61.6  | 66.6  | 71.6  | 75.8  | 77.8  | 77.3  | 74.4  | 69.2  | 62.4  | 58.9 |
|                                                                               |             | MCDB   | 59.3  | 62.9  | 68.8  | 75.7  | 81.0  | 86.2  | 87.2  | 86.2  | 81.0  | 75.2  | 66.8  | 62.9 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 20.5  | 22.1  | 23.3  | 23.3  | 21.3  | 19.7  | 18.6  | 18.6  | 18.6  | 22.0  | 23.2  | 20.8 |
|                                                                               |             | MCDBR  | 25.9  | 27.5  | 27.4  | 26.0  | 24.4  | 23.1  | 21.1  | 21.8  | 21.9  | 24.2  | 25.9  | 24.4 |
|                                                                               |             | MCWBR  | 18.0  | 17.6  | 14.1  | 10.8  | 8.7   | 7.0   | 6.8   | 6.7   | 8.4   | 11.6  | 15.8  | 16.9 |
|                                                                               | 5% WB       | MCDBR  | 22.5  | 24.9  | 22.3  | 21.1  | 20.1  | 20.2  | 18.9  | 18.7  | 17.2  | 17.9  | 20.0  | 21.4 |
|                                                                               |             | MCWBR  | 17.9  | 18.6  | 14.2  | 10.7  | 8.3   | 7.2   | 7.3   | 6.7   | 7.5   | 10.3  | 15.0  | 17.0 |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.321  | 0.335 | 0.369 | 0.408 | 0.464 | 0.501 | 0.549 | 0.525 | 0.431 | 0.392 | 0.341 | 0.319 |      |
|                                                                               | taud        | 2.481  | 2.436 | 2.374 | 2.263 | 2.150 | 2.088 | 1.973 | 2.029 | 2.288 | 2.363 | 2.467 | 2.523 |      |
|                                                                               | Ebn at Noon | 280    | 288   | 286   | 278   | 263   | 252   | 239   | 243   | 262   | 263   | 269   | 274   |      |
|                                                                               | Edh at Noon | 27     | 31    | 36    | 43    | 49    | 51    | 57    | 53    | 39    | 33    | 27    | 24    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 838    | 1048  | 1387  | 1741  | 1915  | 2024  | 1921  | 1735  | 1466  | 1199  | 920   | 729   |      |
|                                                                               | RadStd      | 53     | 119   | 101   | 120   | 149   | 138   | 119   | 87    | 122   | 153   | 74    | 61    |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (68 neighbors) | +0.83 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +286  | +205  |

Nomenclature: See separate page